

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Dulles corridor, Virginia -- station KIAD, America/New_York
Committed pair OSM 693381107 -> 545396372
Amazon IAD81 / Amazon IAD62
Facade gap 137.7 m between the two halls
Weather record 43,763 real hourly records from KIAD
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3593 C at 265 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs.
Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2025-03-11 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 11 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.337	18.0	4.135	1.057
01	mechanical	yes	switch budget	4.337	18.0	4.135	0.998
02	mechanical	yes	switch budget	4.057	18.0	3.025	1.167
03	mechanical	yes	switch budget	2.924	18.0	1.188	1.024
04	mechanical	yes	switch budget	2.952	18.0	0.805	0.996
05	mechanical	yes	switch budget	2.367	18.0	0.245	1.172
06	mechanical	yes	switch budget	1.007	18.0	-0.865	0.987
07	mechanical	yes	switch budget	2.672	18.0	0.245	1.640
08	mechanical	yes	switch budget	6.232	18.0	5.579	1.913
09	mechanical	yes	switch budget	9.979	18.0	11.443	2.362
10	mechanical	yes	switch budget	11.935	18.0	14.216	2.258
11	mechanical	yes	switch budget	15.601	18.0	17.799	1.801
12	mechanical	no	dry-bulb	18.607	18.0	19.564	1.371
13	mechanical	no	dry-bulb	19.909	18.0	21.364	1.176
14	mechanical	no	dry-bulb	22.224	18.0	22.023	1.297
15	mechanical	no	dry-bulb	22.631	18.0	23.103	1.073
16	mechanical	no	dry-bulb	23.306	18.0	23.103	1.147

17	mechanical	no	dry-bulb	22.878	18.0	22.543	1.420
18	mechanical	no	dry-bulb	22.483	18.0	21.219	1.333
19	mechanical	no	dry-bulb	20.949	18.0	19.014	1.421
20	mechanical	no	dry-bulb	20.185	18.0	17.799	1.563
21	mechanical	no	dry-bulb	18.833	18.0	15.859	1.694
22	FREE	yes	-	16.069	18.0	12.473	1.377
23	FREE	yes	-	15.818	18.0	13.362	1.140

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.337 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.337 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.057 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.924 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.952 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.367 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.007 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.672 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.232 C against a 18.0 C

limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.979 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.935 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.601 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.607 C against a 18.0 C limit, so it fails by 0.607 C. It would take a limit 0.607 C higher, or a bound 0.607 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.909 C against a 18.0 C limit, so it fails by 1.909 C. It would take a limit 1.909 C higher, or a bound 1.909 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.225 C against a 18.0 C limit, so it fails by 4.225 C. It would take a limit 4.225 C higher, or a bound 4.225 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.631 C against a 18.0 C limit, so it fails by 4.631 C. It would take a limit 4.631 C higher, or a bound 4.631 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.306 C against a 18.0 C limit, so it fails by 5.306 C. It would take a limit 5.306 C higher, or a bound 5.306 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.878 C against a 18.0 C limit, so it fails by 4.878 C. It would take a limit 4.878 C higher, or a bound 4.878 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.483 C against a 18.0 C limit, so it fails by 4.483 C. It would take a limit 4.483 C higher, or a bound 4.483 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.949 C against a 18.0 C limit, so it fails by 2.949 C. It would take a limit 2.949 C higher, or a bound 2.949 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.185 C against a 18.0 C limit, so it fails by 2.185 C. It would take a limit 2.185 C higher, or a bound 2.185 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.833 C against a 18.0 C limit, so it fails by 0.833 C. It would take a limit 0.833 C higher, or a bound 0.833 C tighter, to change this hour.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.069 C, which is 1.931 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.377 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.1882 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.818 C, which is 2.182 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.140 C, and the margin is measured, not

chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.1872 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

WHAT IT IS WORTH OVER FIVE REAL YEARS, AT THIS SITE

Held-out days simulated 913 -- the agent never calibrates on a day it is scored on
Free cooling delivered 14.72 h/day, i.e. 5,376 h/yr
Chiller-hours avoided +401.7 h/yr against a tuned reactive incumbent
12-hour plan stability 94.0 % of 21,879 re-plans change nothing at all

The ladder, one constraint at a time:

N-56-like: notice 0, skill 1.00, no constraint	+71.2 h/yr
+ switch budget 2, min dwell 3 h	+96.4 h/yr
+ dew-point gate 15 C (Green Grid WP#46 p.6)	+124.4 h/yr
+ notice 3 h, skill 0.50 (no perfect forecast)	+401.7 h/yr
+ unanchored, 4 measured FG offsets rotated	-156.0 h/yr

PRICED, IN THIS STATE'S OWN ELECTRICITY TARIFF

Chiller power 156.1 kW per MW of IT load (0.549 kW/ton, the ASHRAE 90.1-2019 minimum)
Electricity 8.72 cents/kWh (Virginia commercial, 2024 annual)
Energy avoided 62,701 kWh per MW of IT load per year
Value \$5,467 per MW of IT load per year

Everything above is PER MEGAWATT OF IT LOAD. This project has never measured a data centre's size and will not invent one; a reader who knows their own IT load multiplies once.

WHAT IS NOT CLAIMED

- The chiller COMPRESSOR only. Fans, chilled-water pumps, condenser pumps and cooling-tower fans keep running, and an airside economizer moves MORE air, so fan power can rise. The unmeasured term has the OPPOSITE SIGN, so this is an upper bound.
- Code-minimum chiller efficiency is the OPTIMISTIC end: Standard 90.1 is a floor, hyperscale plants beat it, and a better chiller saves less per hour switched off.
- Full-load kW/ton OVERSTATES the draw at the moment free cooling is available, because free cooling happens when it is cool and a chiller at a cool condenser runs below its rating. IPLV is the lower published number but is an annual-weighted metric under AHRI's assumed load profile, not the part-load point free-cooling weather sits at.
- The heat rejected is APPROXIMATED by the IT load. UPS, PDU and lighting losses inside the envelope add to it; some overhead sits outside the cooled space, so scaling by full PUE would overshoot. No PUE multiplier is applied.
- EIA STATE-SECTOR AVERAGES, not the site's tariff. A Loudoun County campus buys on a large-general-service contract that is not public.
- No carbon, no water, no maintenance and no demand charges. Tower water use moves the OTHER way when the chiller runs less.
- Every caveat on the hours is inherited: the headline is conditional on the bank sitting on the long facade (the refusal guard is priced at -3,124 h/yr where it fires) and the unanchored row is NEGATIVE. Those rows appear here with their signs intact.

HOW TO REPRODUCE EVERY NUMBER IN THIS REPORT

```
cd INTAKE-ARBITER/src && python run_all.py
```

That rebuilds every artefact from saved data and then audits it: 39 checks, 68 published figures re-read out of the files the code itself wrote, and it exits non-zero on any failure. It makes ZERO API calls -- every input is a saved FortyGuard response, a committed geometry file, or the station's own hourly record.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.